BSCS FINAL PROJECT
Software Requirements Specification

Green Guard

Product Owner

Muhammad Zulkifl Hasan

Presented by:
Group ID: S26CS060

Student Reg#	Student Name
L1S23BSCS0056	Minahil Kamran
L1S23BSCS0070	Hamia Khalil
L1S23BSCS0090	Laiba Farooq

Faculty of Information Technology & Computer Science

University of Central Punjab

Software Requirements Specification

Version: 1

Product Owner:

Muhammad Zulkifl Hasan

Scrum Master

Minahil Kamran

Group S26CS060

Team

Member Name	Roles
Minahil Kamran	Scrum Master, AI Developer, Frontend Developer
Hamia Khalil	Full Stack and AI Developer
Laiba Farooq	Documentation and Cloud Deployment Lead

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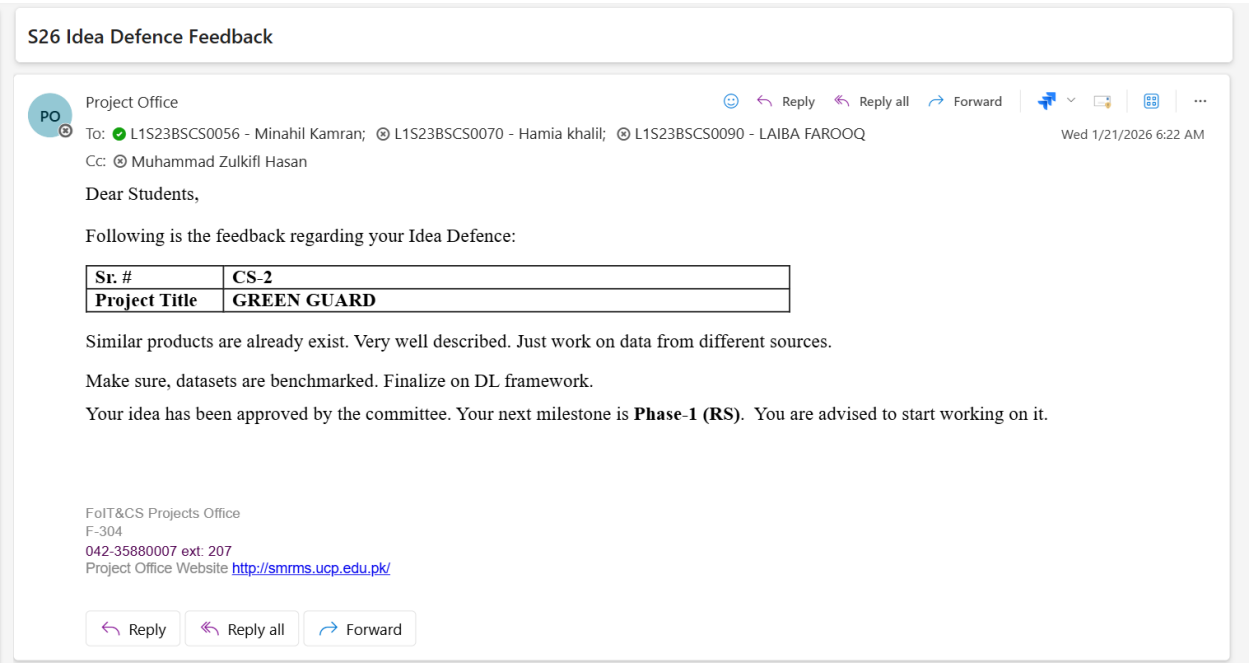
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Revision History & Sprint Log

Version	Sprint	Date	Changes	Owner
1.0	Sprint 1	March 2026	• Requirement gathering • Literature review • Problem analysis • Project objectives defined	Entire Team
1.1	Sprint 2	March 2026	• Research on multimodal deep learning • Research on architectures and agricultural AI systems	Entire Team
1.2	Sprint 3	April 2026	• System architecture finalized • Workflow design completed • Tech stack selected	Minahil Kamran, Hamia Khalil
1.3	Sprint 4	April 2026	• Initial prototyping • SRS documentation	Minahil Kamran, Laiba Farooq
1.4	Sprint 5	May 2026	• Dataset sources identified • Dataset collection initiated (plant disease images, environmental sensor data, temporal data)	Hamia Khalil, Laiba Farooq

Previous Phases Feedback

Idea Defense Feedback (Screenshot)



S26 Idea Defence Feedback

PO

Project Office

Reply | Reply all | Forward

To: L1S23BSCS0056 - Minahil Kamran; L1S23BSCS0070 - Hamia khalil; L1S23BSCS0090 - LAIBA FAROOQ

Wed 1/21/2026 6:22 AM

Cc: Muhammad Zulkifl Hasan

Dear Students,

Following is the feedback regarding your Idea Defence:

Sr. #	**CS-2**
Project Title	**GREEN GUARD**

Similar products are already exist. Very well described. Just work on data from different sources.

Make sure, datasets are benchmarked. Finalize on DL framework.

Your idea has been approved by the committee. Your next milestone is **Phase-1 (RS)**. You are advised to start working on it.

FoIT&CS Projects Office
F-304
042-35880007 ext: 207
Project Office Website http://smrms.ucp.edu.pk/

Reply | Reply all | Forward

Figure 1: Idea Defense Feedback

Abstract

Green Guard is an intelligent plant health monitoring system to help farmers identify plant problems more accurately using deep learning and feature level fusion techniques. Traditional plant disease detection systems mainly rely on leaf images only and often ignore important factors like environmental conditions and historical plant data. Because of this, they may not provide complete or accurate results. The proposed system solves this issue by combining three different types of data like plant leaf images, environmental data and temporal plant data.

The system uses deep learning models to analyze each type of data separately. A Convolutional Neural Network (CNN) model extracts visual features from leaf images, an Artificial Neural Network (ANN) model processes environmental information such as temperature, humidity, soil moisture etc while a Recurrent Neural Network (RNN) model studies historical trends and changes over time. These extracted features are then combined using a feature fusion layer to create a complete understanding of plant health. The fused features are then processed using an ensemble of Multi Layer Perceptron (MLP) models to simultaneously detect plant diseases, nutrient deficiencies and environmental stress conditions.

The project operates in the agricultural domain where biological and environmental factors define the problem context, while all data processing, modeling and prediction are performed using computer science and AI based methodologies.

The main goal of Green Guard is to provide a smart, accurate, and reliable plant monitoring system for modern farming. The system will also generate useful recommendations for farmers like improving irrigation, applying fertilizers or controlling diseases. This project contributes to the fields of artificial intelligence, smart agriculture, and precision farming by introducing a multimodal and multi task learning framework for better plant health diagnosis.

1. Introduction and Background

1.1 Product Vision & Problem Context

Problem Description:

Agriculture plays a major role in food production and the economy, but farmers often face serious challenges in identifying plant diseases, nutrient deficiencies, and environmental stress conditions at an early stage.[2] Traditional methods of crop monitoring mostly depend on manual inspection by farmers or agricultural experts, which can be time consuming, expensive, and inaccurate. [12]

Traditional observation and experience are widely used in agricultural practices, leading to incorrect decisions in many cases, without the presence of smart systems. For instance, plant diseases are typically only detected once they become noticeable on leaves rather than through an automated system. Fertilizer application is frequently based on guesswork and not good soil nutrient analysis, which results in over or under usage. Likewise, irrigation is applied on a fixed schedule or based on intuition rather than actual soil moistures data, that can result in water waste or crop stress. Farmers are also unable to distinguish similar plant diseases without expert assistance during their work. These constraints emphasize the need for intelligent solution, such as Green Guard.

Literature Survey:

Deep learning (DL) has emerged as a powerful approach in modern agriculture by enabling accurate analysis of complex image and environmental data, supporting applications such as crop yield prediction, weed detection, and plant disease identification [6].

Convolutional Neural Networks (CNNs) are widely used for analyzing leaf and crop images to detect diseases, pests, and nutrient deficiencies by extracting spatial features efficiently [9], while Recurrent Neural Networks (RNNs) model time series agricultural data including environmental changes and sensor readings to predict stress conditions over time [7]. Artificial Neural Networks (ANNs) process structured data such as soil nutrients and weather parameters to identify hidden patterns affecting plant health [8]. Ensemble learning integrates CNNs, RNNs, and ANNs to enhance prediction accuracy, robustness, and generalization by combining visual and non-visual data for more reliable and actionable crop monitoring systems [10].

Previous research applied deep learning for early leaf spot disease detection using CNN for feature extraction and RNN based for classification on the *PlantVillage* dataset (nearly 20,000 images). Results showed based models achieved higher accuracy than traditional machine learning methods [11]. This study evaluated only a single CNN architecture, relied solely on image data, and did not compare multiple deep learning or ensemble approaches.

Another study proposed an ensemble framework combining CNNs and RNNs to analyze leaf images with time series environmental data. Ensemble techniques such as majority voting and weighted averaging improved prediction accuracy on the *Kaggle Cotton* dataset [12].

A separate study used CNNs to learn unsupervised features for 44 plant species and applied deconvolutional visualization for feature interpretation. CNN based features outperformed traditional handcrafted techniques [13]. The research focused only on species identification, not on disease detection, nutrient deficiency, or integrated plant health monitoring using ensemble models.

Another work developed a cotton leaf classification system using Random Forest, SVM, and ensemble learning. The ensemble approach improved accuracy compared to individual classifiers [14].

Existing Work	Focus Area	Limitations Identified	Literature Gap	How This Project Addresses the Gap
Plant leaf disease detection using a CNN LSTM [2]	Disease detection	Uses only a single CNN LSTM architecture and image dataset	No usage of multiple models and non image data	Uses multiple models and combines image and sensor data
CNN based plant species identification [3]	Plant species identification	Focuses on species recognition only	No disease detection, plant health monitoring or usage of multiple models	Performs disease, nutrient, and environment monitoring and multimodal
Cotton leaf classification using ML [4]	Cotton leaf classification	Uses traditional machine learning	No deep learning based multimodal and focus on only one specie	Uses deep learning models with multimodals and works on multiple species
Image Processing based Crop Disease Detection	Disease identification	Uses traditional image processing techniques	Outdated methods with lower accuracy	Uses deep learning models for improved accuracy
IoT based Smart Agriculture System	Environmental monitoring	Focuses only on sensor data	No integration with image based disease detection	Combines image data with sensor data
Single Crop Monitoring Systems [1]	Crop specific monitoring	Designed for one crop only	Not works on multispecies	Designed to support multiple plant species

Table 1: Comprehensive Literature Gap

Proposed Solution:

Green Guard is a proposed multi modal, multi task deep learning system, which combines several Artificial Intelligence methods for the monitoring of plant health. It employs a type of neural network called a convolutional neural network (CNN) to extract visual features from leaf images, such as spots, lesions, discoloration, and disease textures. Structured environmental and sensor data, such as temperature, humidity, soil moisture, pH and nutrient levels are processed by Artificial Neural Networks (ANNs) to discover hidden patterns that influence plant health. In the case of sequential and historical data, Recurrent Neural Networks (RNNs) models are used to model the temporal dependencies and environmental trends over time.

Independent feature extraction is followed by a feature level fusion technique in which the image feature vector, environmental and temporal feature vectors are concatenated and then input to a dense fusion layer to learn cross modal interaction and adaptively identify feature importance. An ensemble learning approach is then adopted, which is composed of two MLP based models (shallow and deep) to enhance the robustness and generalization of the system by inputting the same fused representation.

Finally, the results of both models are fused by using the average voting ensemble method to get stable predictions for several tasks, such as disease detection, nutrient deficiency identification and stress analysis. The system can also do multi task learning with multiple output heads, allowing prediction of multiple types of agricultural conditions from the same representation.

The system is developed with a focus on user friendly and low literacy friendly interface, using simple visual indicators, icons and color coded alerts as the mediums. This means that even semi literate or illiterate farmers can be able to understand the status of the plants, results of the disease detection and what needs to be done without having to be technical.

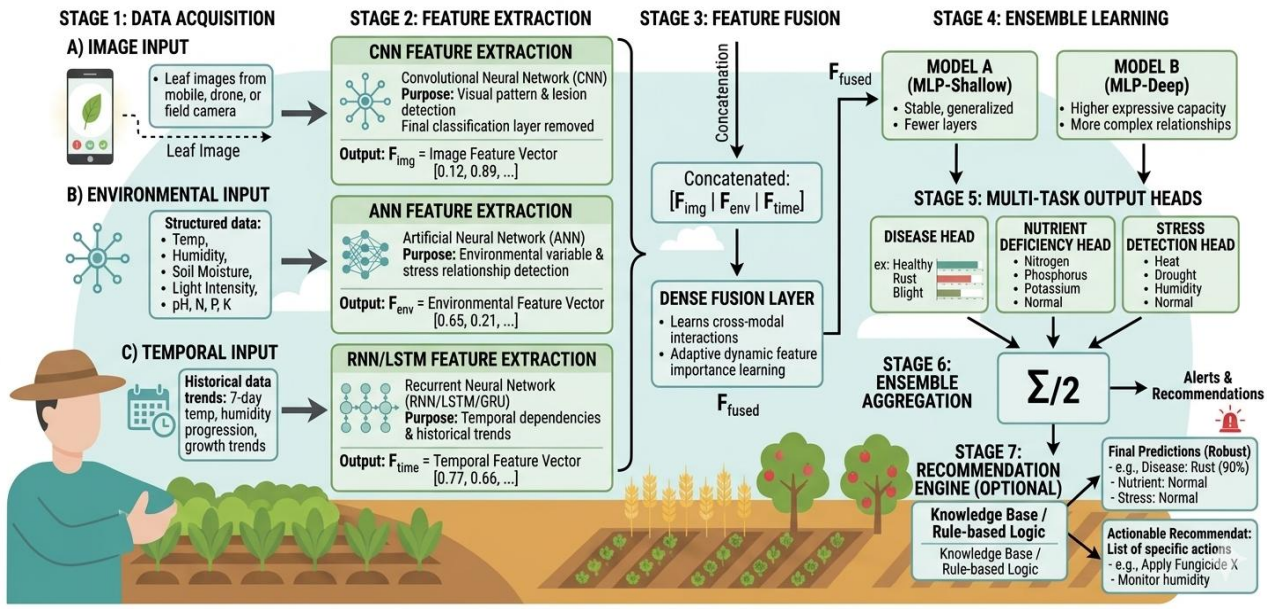


Figure 2: Proposed System Workflow

Target Users:

- **Farmers:** Farmers are the main users of the system. They can monitor crop health, detect diseases and stress conditions early, and make better farming decisions to reduce crop losses and improve productivity.
- **Agricultural Experts:** Agricultural experts can use the system to analyze plant health conditions and provide better guidance and recommendations for crop management.
- **Greenhouse Managers:** Greenhouse managers can monitor environmental conditions such as temperature, humidity, and soil moisture to maintain healthy crop growth inside controlled environments.
- **Agricultural Researchers:** Researchers can use the system for studies related to smart farming, plant disease detection, and AI based agricultural monitoring.
- **Agriculture Companies:** Agriculture companies can integrate the system into smart farming solutions and automated crop monitoring platforms.
- **Agricultural Consultants:** Agricultural consultants can use the system to identify crop issues and suggest suitable treatments and farming practices to farmers.
- **System Administrator:** The system administrator will manage system operations, user access, datasets, and overall platform maintenance.

Objectives of the System:

The main objectives of Green Guard are:

- To detect plant diseases using leaf images.
- To identify nutrient deficiencies in crops.
- To detect environmental stress conditions affecting plant growth.
- To combine image, sensor, and historical data for improved accuracy.
- To provide intelligent recommendations for farmers.

- To generate biologically consistent predictions through AI based analysis and agricultural knowledge validation.
- To support smart and precision farming practices using AI technologies.

User Research:

Observation of Current Agricultural Process at Dogar Farm:

To understand how crop monitoring and plant health management work in the real world, our FYP team conducted a field observation study. Since our project, Green
Guard, aims to solve actual practical issues using technology, we needed to see exactly how small to medium scale farmers manage their crops, notice diseases, and keep track of environmental conditions without automated systems.

Field Details:

- **Location: "**Dogar Farms" near railway station Kasur
- **Type:** Typical semi commercial small to medium scale farm environment.
- **Crops Observed:** Tomatoes, potatoes, and green leafy vegetables
- **Participants:** Farm supervisor (15+ years experience) , field workers
- **Observation Timing:** Early morning farming activities, when crop inspection and field work are most active.
- **Duration:** Several hours of routine farm operations (crop checking, watering, and maintenance tasks).

Scenario and Ground Observations:

During our time at the farm, we walked alongside the workers to see their daily routine. The process of checking crop health was entirely manual. The laborers walked slowly between the crop rows, looking down at random plants to see if anything looked unusual. If a leaf appeared yellow, spotted, or wilted, they would stop to inspect it closer.



Figure 3: Similar looking diseases

Early Blight (left) and Late Blight (right) on leaves of tomato crop, showing overlapping visual symptoms.

While discussing their work, the farm supervisor explained that they do not have any scientific tools to test the soil or environment on a daily basis. To check if the soil has enough water, the workers simply dig up a tiny bit of mud with their hands or sticks, look at its color, and feel the texture to guess if it is dry or damp.



Figure 4: Manual Soil Check

Air temperature and humidity are judged completely by personal physical feeling or by checking general public weather apps on their phones, which only provide broad city level data rather than precise micro climate conditions for their specific fields.

We noticed a clear example of manual limitations when the supervisor pointed out a patch of potato tubers that looked diseased compared to other patch of healthy potatoes. A discussion started between the supervisor and a laborer about whether they were drying up due to intense morning heat, or if they were suffering from an early stage fungal infection.



Figure 5: Healthy Vs Diseased Potatoes

Because they lacked a clear way to verify the issue, they decided to wait a couple of days and apply extra water, hoping it would fix itself. This trial and error method highlights how vulnerable traditional farming is when dealing with early stage plant stress

.



Figure 6: Healthy Crops Corn Plant (left) Tomato (right)

Identified Outcomes

Based on our observations and detailed conversations with the farm staff, we identified several critical problems that directly impact crop yield and increase financial risks for farmers:

- Inefficient and Delayed Manual Disease Detection
- Extreme Difficulty in Early Symptom Identification
- Complete Lack of Micro Environmental Monitoring
- Absence of Smart Agricultural Assistive Systems

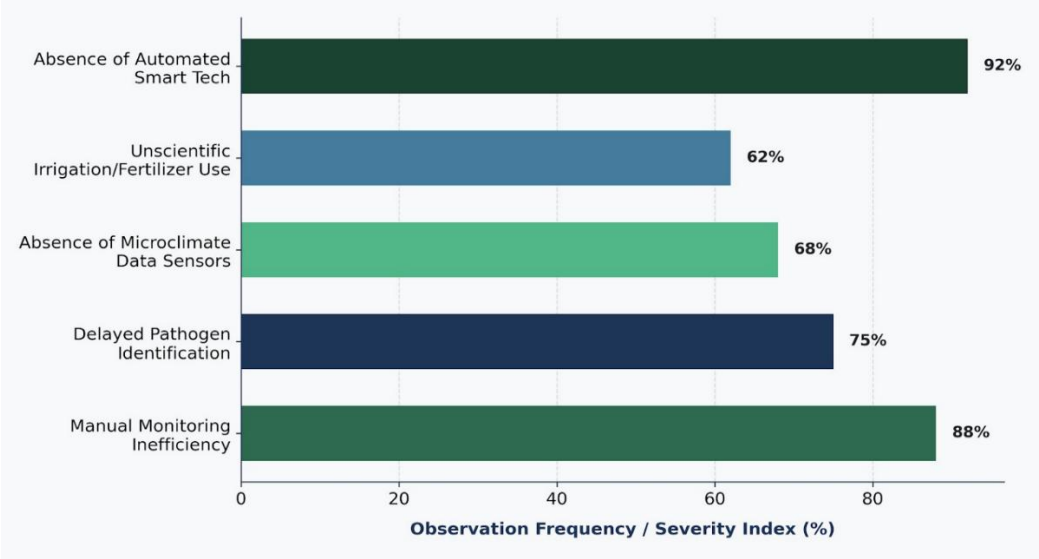


Figure 7: Identified Field Weaknesses

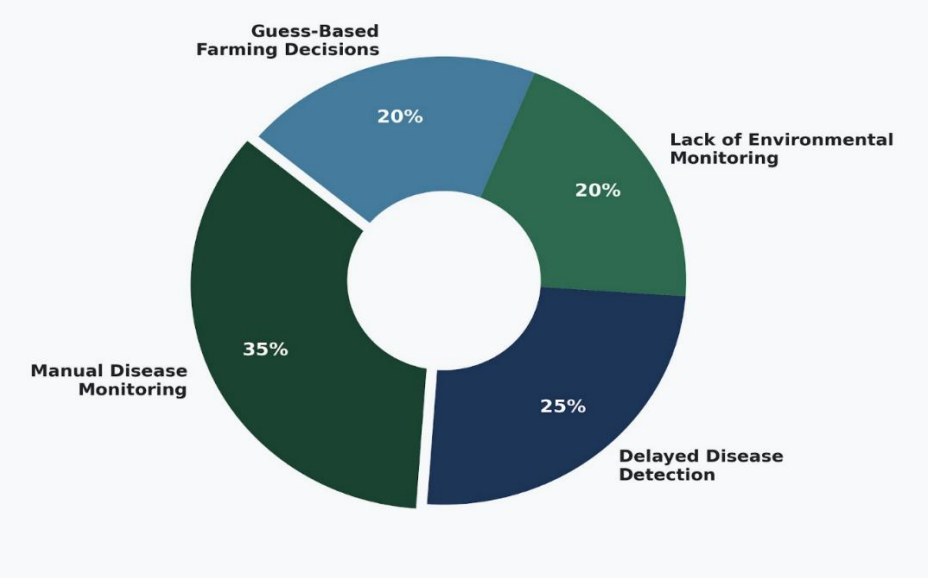


Figure 8: Distribution of Observed Agricultural Challenges

Observation of Solution at FYP Student Led Research & Innovation Event:

The observation phase was conducted during participation in the FYP Student Led Research & Innovation event, where different agriculture related monitoring challenges, smart farming ideas, and practical crop management issues were discussed and analyzed. During this activity, observations were made regarding the limitations of traditional plant monitoring approaches and the increasing need for intelligent agricultural systems.

The observation revealed several important issues faced in agricultural environments:

- Farmers mainly rely on manual inspection for detecting plant diseases and stress conditions.
- Early symptoms of nutrient deficiencies and environmental stress are often difficult to identify visually.
- Environmental factors such as temperature, humidity, and soil moisture significantly affect plant health but are rarely analyzed together with visual symptoms.
- Existing systems mostly focus only on image based disease detection and ignore multimodal agricultural data.
- Lack of intelligent monitoring systems may lead to delayed diagnosis, reduced crop quality, and productivity loss.

These observations helped define the core problem addressed by Green Guard and supported the development of a multimodal AI based agricultural monitoring framework.

The proposed system received encouraging feedback for its multimodal approach and relevance to smart agriculture multimodal learning. The project also received appreciation for addressing real world agricultural challenges through computational techniques.

University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

Ref No.: UCP/ORIC/FYP/2026/01

Date: May 05, 2026

Supervisor 1	Supervisor 2	Team Lead	Group Members
Mr. Muhammad Zulkifl Hassan, Principal Lecturer Faculty of Information Technology & Computer Sciences	Dr. Sadaf Hameed Associate Professor Faculty of Science and Technology	Minahil Kamran L1S23BSCS0056	Hamia Khalil L1S23BSCS0070 Laiba Farooq L1S23BSCS0090 Sabra Anwar L1S23BSBT0022

Subject: Award of FYP: Student-Led Research & Innovation Fund

Dear Team Members,

I am pleased to inform you that the University of Central Punjab (UCP) has provisionally approved your Final Year Project (FPY) titled **"GreenGuard"** under the FYP-Student-Led Research & Innovation Fund. The project has been awarded a total funding of Rs. 50000/- (Fifty Thousand Rupees Only) for a maximum duration of 12 months.

The approved budget allocation is as follows:

S. No.	Item	Estimated Cost (PKR)
1	External Cloud GPU Support Primary: AWS GPU (g4dn.xlarge) Type: NVIDIA T4Pay as you go model	13,000
2	Logitech C920 HD Pro Webcam	24,500
3	Environmental Sensor Kit BME 680	4,076
4	External Storage (SSD 256GB)	5,300
5	Arduino Uno R3 SMD (Sensor Integration)	1,400
6	Miscellaneous (Cables, Setup, Documentation)	1,724
	Total	50,000

All financial matters, including purchases, payments, and utilization of project funds, shall be managed through the Office of Research, Innovation and Commercialization (ORIC). Purchase memos and payment requests must be processed through ORIC and routed to the Treasurer's Office for financial approval and disbursement, as per UCP financial policies and procedures.

Please ensure that all project activities, expenditures, and reporting comply with the guidelines and timelines. ORIC will provide periodic monitoring and support throughout the project duration.

Congratulations on your successful award. We look forward to your continued contribution to advancing research, innovation, and academic excellence at UCP.
With best regards,

Prof. Dr. Muhammad Akhyar Farrukh
Director, ORIC
Office of Research, Innovation & Commercialization
University of Central Punjab

cc:
1. Prorector
2. Concerned Dean
3. Treasurer
4. Manager ORIC
5. Team Members
6. Supervisors (I and II)

Khayabane-e-Jinnah, Avenue-1, Johar Town, Lahore, Pakistan. Tel: 92-42-35880007 Ext: 343/671/709
E-mails: oric@ucp.edu.pk, dir.oric@ucp.edu.pk Website: www.ucp.edu.pk

Figure 9: Observation Appreciation and Award Letter

Expected Benefits:

Expected Outcome	**Potential Beneficiaries**
Early detection of plant diseases	Small scale farmers, Commercial farming industries, Government agricultural departments
Improved crop monitoring and management	Commercial farming industries, Farming organizations, Greenhouse managers
Better understanding of environmental effects on crops	Agricultural researchers, Farming organizations, Smart agriculture startups
Reduced crop losses	Small scale farmers, Commercial farming industries, Government agricultural departments
Increased farming efficiency	Commercial farming industries,Farming organizations, Smart agriculture startups
Support for data driven agriculture	Agricultural researchers, Smart agriculture startups, Precision farming organizations
Assistance in agricultural decision making	Farmers, Agricultural consultants, Government agricultural departments
Improved reliability of plant health monitoring through AI driven analysis and agricultural knowledge based validation	Agricultural researchers, Smart agriculture startups, Precision farming organizations

1.2 Stakeholders Identification

The following stakeholders are involved in or affected by the proposed Green Guard system. Each stakeholder has different roles, responsibilities, and expectations from the system.

Stakeholder	**Role**	**Expectations**	**Priority**
Farmers	Primary End Users	Early disease detection, easy monitoring, better crop management	High
Agricultural Experts	Field Advisors	Accurate analysis, environmental insights, reliable predictions	High

Project Supervisor (Muhammad Zulkifl Hasan)	Project Supervisor	Proper research methodology, technical quality, successful completion	High
Development Team	System Developers	Clear requirements, manageable tasks, smooth integration	High
Government Agricultural Departments	Policy & Support Stakeholders	Improved agricultural productivity and smart farming support	Medium
Investors	Funding Stakeholders	Innovative solution, scalability, practical impact	Medium
Environmental Agencies	Environmental Observers	Sustainable farming support and environmental monitoring	Low
Farming Communities	Indirect Beneficiaries	Better farming awareness, improved crop quality, reduced losses	Medium

2. Agile Framework Setup

The Green Guard project will be developed in an iterative fashion, with effective collaboration and continuous improvement, by following the Agile software development methodology as outlined below in the Scrum framework

The project will be broken up in small sprints of development, with each sprint dealing with a functional piece of the system, such as pre processing the datasets, identifying diseases, analyzing the environment, building the dashboard, or generating alerts. All epics, user stories, and tasks will be kept in the team's product backlog. Sprint planning meetings will be held for the purpose of choosing tasks, establishing sprint goals, and assigning sprint tasks. Weekly Scrums will be used to monitor progress, review what has been accomplished, clarify issues encountered, and enhance team coordination.

Sprint review and sprint sessions will take place after each sprint to discuss and reflect on what has been accomplished, find areas for feedback and input, and plan for the next iteration. Backlog management, sprint tracking, Kanban Boards, task organization will be achieved using agile project management tools like Notion The goal of the team is to continuously test, deliver the system on time, and be flexible by applying Agile and Scrum practices.

Agile Team Roles:

Team Roles	**Names**	**Responsibilities**
Product Owner	Muhammad Zulkifl Hasan	• Defining project requirements • Reviewing sprint outcomes • Providing domain and research guidance
Scrum Master	Minahil Kamran	• Managing sprint planning • Monitoring task progress • Conducting team meetings
Development Team	Hamia Khalil, Minahil Kamran, Laiba Farooq	• Data collection • Backend and frontend implementation • DL model development • System integration • Testing and evaluation • Documentation preparation

Generalized Sprint Planning:

The Green Guard project will follow a structured Agile Scrum based development methodology distributed across weekly and sprint based milestones from March 2026 to January 2027. The project duration is divided into 20 sprints each with a 2 week duration. Each sprint focuses on a specific module or set of functionalities.

Sprint No.	Timeline	Duration	Generalized Sprint Activities
Sprint 1	Week 1–2 (March 2026)	2 Weeks	• Requirement gathering • Problem analysis • Literature review • Identification of project objectives
Sprint 2	Week 3–4 (March 2026)	2 Weeks	• Research on multimodal deep learning architectures • Research on agricultural AI systems, and related studies
Sprint 3	Week 5–6 (April 2026)	2 Weeks	• Finalization of system architecture • Workflow design • Technology stack selection • Project scope refinement
Sprint 4	Week 7–8 (April 2026)	2 Weeks	• Initial system prototyping • UI wireframing • Dataset source identification
Sprint 5	Week 9–10 (May 2026)	2 Weeks	• Dataset collection for plant disease images • Environmental sensor readings • Temporal agricultural data
Sprint 6	Week 11–12 (May 2026)	2 Weeks	Dataset cleaning, preprocessing, normalization, augmentation, and labeling
Sprint 7	Week 13–14 (June 2026)	2 Weeks	Development of CNN based image feature extraction and disease classification module
Sprint 8	Week 15–16 (June 2026)	2 Weeks	Training, validation, optimization, and performance tuning of CNN architecture

Sprint 9	Week 17–18 (July 2026)	2 Weeks	Development of ANN based environmental and sensor data processing module
Sprint 10	Week 19–20 (July 2026)	2 Weeks	Development of RNN based temporal trend analysis and sequential data processing module
Sprint 11	Week 21–22 (August 2026)	2 Weeks	Integration of CNN, ANN, and RNN feature extraction pipelines into a unified framework
Sprint 12	Week 23–24 (August 2026)	2 Weeks	Implementation of feature concatenation, dense feature fusion layers, and multimodal representation learning
Sprint 13	Week 25–26 (September 2026)	2 Weeks	Development of Ensemble Model A using shallow MLP architecture and prediction aggregation
Sprint 14	Week 27–28 (September 2026)	2 Weeks	Development of Ensemble Model B using deep MLP architecture and ensemble optimization
Sprint 15	Week 29–30 (October 2026)	2 Weeks	Implementation of average voting ensemble mechanism.
Sprint 16	Week 31–32 (October 2026)	2 Weeks	Development of disease detection, nutrient deficiency prediction, stress analysis heads, and recommendation engine
Sprint 17	Week 33–34 (November 2026)	2 Weeks	• Frontend dashboard development • Visualization modules • API integration
Sprint 18	Week 35–36 (November 2026)	2 Weeks	• Model evaluation • Confusion matrix analysis • Ablation studies

Sprint 19	Week 35–36 (December 2026)	2 Weeks	User interface enhancement and deployment preparation
Sprint 20	Week 35–36 (December 2026)	2 Weeks	Report writing and final documentation

Agile Workflow for Green Guard:

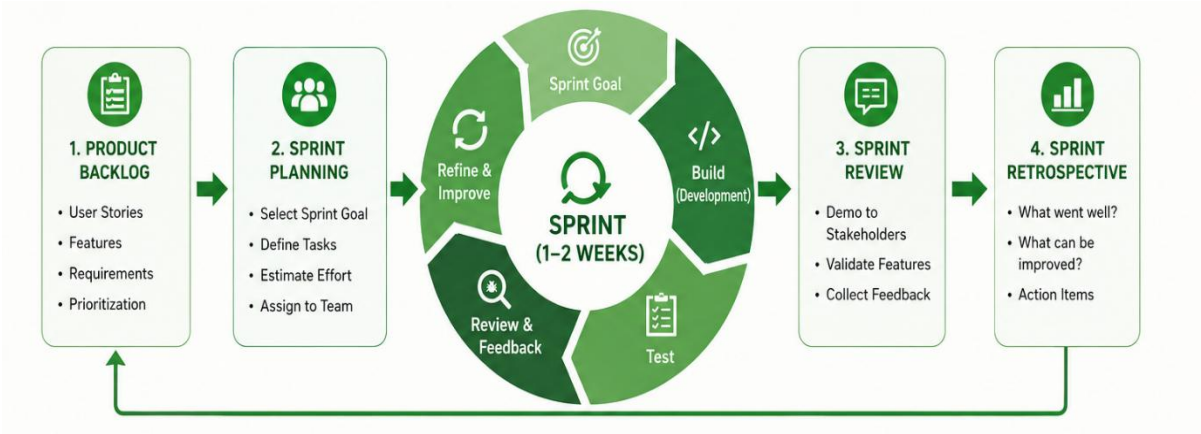


Figure 10: Agile Workflow Green Guard

2.1 Epics For the Project

An Epic is a major functional module of the Green Guard system, such as disease detection, data processing, or recommendation generation. It represents a high level feature that is further broken into smaller tasks for implementation.

Table of Epics

Epic ID	Epic Name	Description	Related Problem Area
EP-01	User Authentication & Management	Handles user registration, login, and authentication for secure system access control	Lack of secure system access
EP-02	Image Acquisition & Processing	Performs image preprocessing, resizing, normalization, augmentation, and enhancement for plant disease detection	Raw agricultural images are noisy and inconsistent
EP-03	Dataset Management	Handles collection, labeling, cleaning, and organization of multimodal datasets	Poor quality and unstructured agricultural data

EP-04	Data Synchronization & Alignment	Aligns image, environmental, and time-series data into a synchronized multimodal format	Multimodal data inconsistency
EP-05	CNN based Visual Feature Extraction	Extracts visual disease features from plant leaf images using CNN models	Difficulty in detecting plant diseases from raw images
EP-06	ANN based Environmental Processing	Processes environmental and sensor based data such as temperature, humidity, and soil conditions using ANN	Environmental factors are not properly analyzed
EP-07	RNN Temporal Trend Analysis	Captures time series patterns in environmental and crop data using sequential models	Lack of temporal pattern understanding
EP-08	Multimodal Feature Fusion Pipeline	Combines CNN, ANN, and RNN outputs to learn relationships between different data modalities	Weak cross modal learning and lack of integration
EP-09	Ensemble Learning Module	Uses ensemble techniques to improve prediction stability and accuracy	Low model stability and accuracy
EP-10	Plant Health Analysis & Detection	Detects plant diseases, nutrient deficiencies, and environmental stress using fused multimodal AI features	Delayed or inaccurate crop health detection
EP-11	Recommendation Engine	Provides actionable farming advice such as fertilizer, irrigation, and treatment suggestions	Lack of decision support for farmers
EP-12	Backend API & System Integration	Integrates AI models into a unified backend system for real-time inference	Lack of deployable AI system integration
EP-13	Model Training, Optimization & Evaluation	Handles training pipeline, hyperparameter tuning, loss optimization, and model evaluation metrics	Poor model performance and lack of structured evaluation
EP-14	Explainability & Confidence Analysis	Provides prediction confidence scores and interpretable AI outputs	Lack of trust in AI predictions
EP-15	User Interface & Dashboard Visualization	Displays predictions, analytics, graphs, and system insights in an interactive dashboard	Complex AI system not user-friendly
EP-16	Alert & Notification System	Generates real time alerts for diseases, nutrient deficiencies, and environmental risks	Delayed awareness of critical crop conditions
EP-17	Deployment & System Maintenance	Handles system deployment, hosting, monitoring, and model updates	AI systems face deployment and maintenance challenges
EP-18	Security	Ensures secure authentication, data protection, encrypted storage, and API security	User data and system APIs are vulnerable to

			unauthorized access

2.2 Product Backlog

A Backlog is a specific, actionable task within an Epic that contributes to building the Green Guard system. It defines a small development step like training a model, processing data, or building a UI feature.

Product Backlog Table

Backlog ID	Epic	Feature Description	Priority
PB-01	User Authentication & Management	Implement user registration with email and password	High
PB-02	User Authentication & Management	Develop secure login and logout functionality	High
PB-03	User Authentication & Management	Implement password reset and recovery	Medium
PB-04	User Authentication & Management	Create user profile management module	Medium
PB-05	Image Acquisition & Processing	Enable plant image upload from device	High
PB-06	Image Acquisition & Processing	Implement image preprocessing (resizing, normalization, enhancement)	High
PB-07	Image Acquisition & Processing	Apply image augmentation techniques for training	Medium
PB-08	Dataset Management	Collect and organize multimodal agricultural datasets	High
PB-09	Dataset Management	Store image, environmental, and temporal data in database	High
PB-10	Data Synchronization & Alignment	Synchronize image and environmental datasets	High
PB-11	Data Synchronization & Alignment	Align temporal records for sequential analysis	Medium
PB-12	CNN-based Visual Feature Extraction	Develop and train CNN model for plant feature extraction	High
PB-13	ANN-based Environmental Data Processing	Process environmental and weather data using ANN	High

PB-14	RNN-based Temporal Trend Analysis	Analyze temporal crop patterns using RNN models	High
PB-15	Multimodal Feature Fusion Pipeline	Combine image, environmental, and temporal features	High
PB-16	Multimodal Feature Fusion Pipeline	Develop multimodal learning architecture for joint representation learning	High
PB-17	Ensemble Learning Module	Implement ensemble learning with model averaging and optimization	High
PB-18	Plant Health Analysis & Detection	Detect plant diseases, nutrient deficiencies, and environmental stress	High
PB-19	Recommendation Engine	Generate treatment, fertilizer, and irrigation recommendations	High
PB-20	Backend API & System Integration	Develop backend APIs and integrate AI models for inference	High
PB-21	Model Training, Optimization & Evaluation	Train models, tune hyperparameters, and evaluate using performance metrics	High
PB-22	Explainability & Confidence Analysis	Provide prediction confidence scores and explainable AI outputs	Medium
PB-23	User Interface & Dashboard Visualization	Design interactive dashboard for predictions and analytics	High
PB-24	Alert & Notification System	Send real-time alerts for plant diseases and stress conditions	High
PB-25	Deployment & System Maintenance	Deploy system on cloud/server and maintain performance	High
PB-26	Security	Implement authentication, encryption, and secure API access	High

2.3 User Stories

User stories define system requirements from the end user perspective, describing what each user wants to achieve from the Green Guard system. Each story is linked to a backlog item with clear acceptance criteria to ensure proper implementation and validation.

User Story Board

Story ID	Backlog ID	User Story	Priority	Acceptance Criteria	Focal Person
US-01	PB-01	As a user, I want to register an account so that I can access the system	High	Valid email/password creates account; invalid input shows error	Minahil Kamran
US-02	PB-02	As a user, I want to log in so that I can securely access my dashboard	High	Correct credentials allow login; invalid credentials show error	Minahil Kamran
US-03	PB-03	As a user, I want to reset my password so I can recover my account	Medium	Reset email sent and link allows password update	Hamia Khalil
US-04	PB-04	As a user, I want to manage my profile so I can update my information	Medium	Profile data can be viewed and updated successfully	Minahil Kamran
US-05	PB-05	As a user, I want to upload plant images so the system can analyze them	High	Valid images uploaded; invalid formats rejected	Laiba Farooq
US-06	PB-10	As a system, I want to synchronize multimodal datasets for accurate analysis	High	Image, environmental, and temporal data are correctly aligned	Hamia Khalil
US-07	PB-12	As a user, I want to extract visual features using CNN for disease detection	High	CNN successfully extracts meaningful image features	Minahil Kamran, Hamia Khalil
US-08	PB-13	As a user, I want to process environmental data using ANN	High	ANN generates valid environmental feature outputs	Minahil Kamran, Hamia Khalil
US-09	PB-14	As a user, I want to analyze temporal trends using RNN	High	RNN processes sequential data and produces trends	Minahil Kamran, Hamia Khalil

US-10	PB-15	As a user, I want to fuse multimodal features for improved prediction	High	CNN, ANN, RNN outputs are successfully combined	Minahil Kamran, Hamia Khalil, Laiba Farooq
US-11	PB-16	As a developer, I want ensemble learning to improve prediction reliability	High	Ensemble model improves prediction accuracy	Minahil Kamran, Hamia Khalil, Laiba Farooq
US-12	PB-17	As a user, I want to detect plant health issues from multimodal data	High	System detects disease, deficiency, and stress correctly	Minahil Kamran, Hamia Khalil, Laiba Farooq
US-13	PB-19	As a user, I want recommendations to improve crop health	High	System provides actionable farming suggestions	Minahil Kamran, Hamia Khalil
US-14	PB-23	As a user, I want visual analytics so that I can easily understand environmental and prediction trends	High	Graphs and charts display real-time system data without rendering errors	Minahil Kamran
US-15	PB-19	As a user, I want alerts so that I can be notified immediately about crop risks	High	System triggers alerts when disease or stress conditions are detected	Laiba Farooq
US-16	PB-19	As a system, I want backend APIs to connect frontend and AI models	High	APIs return correct responses within acceptable time	Hamia Khalil
US-17	PB-20	As a researcher, I want to evaluate model performance	High	Metrics like accuracy, precision, recall, F1-score generated	Laiba Farooq
US-18	PB-21	As a user, I want to view confidence scores to trust predictions	Medium	Confidence score displayed with each prediction	Laiba Farooq
US-19	PB-22	As a researcher, I want explainable AI outputs for interpretation	Medium	Visual explanations highlight important features	Minahil Kamran,
US-20	PB-23	As a user, I want a dashboard to view analytics and results	High	Dashboard loads predictions and graphs correctly	Minahil Kamran,

US-21	PB-24	As a user, I want visual analytics to understand trends	High	Charts display environmental and prediction trends	Minahil Kamran
US-22	PB-25	As a user, I want alerts for crop risks	High	System triggers alerts for diseases and stress conditions	Hamia Khalil
US-23	PB-26	As a developer, I want deployment setup so the system runs in production	Medium	System is successfully deployed and accessible	Laiba Farooq
US-24	PB-27	As a user, I want secure authentication and data protection	High	Unauthorized access blocked; sessions secured	Minahil Kamran, Hamia Khalil

2.4 Story Points & Estimation

Story point estimation in the Green Guard project is used to measure the effort and complexity of developing AI modules such as disease detection, multimodal fusion, and recommendation systems. It helps the team plan and distribute work across sprints based on the difficulty of each feature. Story points are assigned using values like 1, 2, 3, 5, and 8 to represent increasing levels of difficulty.

Story Point Estimation Table

Story ID	User Story (Short)	Complexity Level	Story Points	Justification
US-01	User Registration	Medium	3	Requires form validation, secure password handling, and database integration
US-02	User Login	Medium	3	Authentication logic with session management and error handling
US-03	Password Reset	Medium	3	Email verification with secure token based reset mechanism
US-04	Profile Management	Low	2	Basic CRUD operations for user profile data
US-05	Image Upload	Medium	5	File handling, validation, and preprocessing pipeline integration
US-06	Data Synchronization	High	8	Complex alignment of image, environmental, and temporal datasets
US-07	CNN Feature Extraction	High	8	Deep learning model design and feature extraction from images

US-08	Environmental Data Processing (ANN)	High	5	ANN design and structured sensor data preprocessing
US-09	Temporal Analysis (RNN)	High	8	Sequential modeling using RNN/ for time series data
US-10	Feature Fusion Pipeline	High	8	Integrates CNN, ANN, and RNN outputs into unified representation
US-11	Ensemble Learning	High	8	Combines multiple models to improve prediction stability and accuracy
US-12	Disease Detection	High	5	Classification based on fused multimodal features
US-13	Nutrient Deficiency Detection	High	5	Multi class classification of nutrient imbalance
US-14	Environmental Stress Detection	High	5	Detection of stress conditions using multimodal inputs
US-15	Recommendation System	High	5	Rule-based and AI-driven farming recommendations
US-16	Backend API Integration	Medium	5	API development and integration with AI models
US-17	Model Evaluation Metrics	Medium	3	Computation of accuracy, precision, recall, F1-score
US-18	Confidence Score Display	Low	2	Simple UI display of prediction confidence values
US-19	Explainable AI Visualization	High	5	Heatmaps and interpretability techniques for model explanation
US-20	Dashboard UI	Medium	5	UI development with charts and analytics integration
US-21	Visual Analytics (Graphs)	Low	3	Chart rendering and environmental trend visualization
US-22	Alert System	Medium	5	Event-based notifications and risk alerts
US-23	Deployment Setup	High	8	Server setup, hosting, and production configuration
US-24	Security & Authentication	High	5	Secure login, authorization, and session management
US-25	Data Encryption	High	5	Encryption of stored and transmitted sensitive data
		Total Points	**112**	

2.5 Sprint Planning & Sprint Backlog

Sprint planning divides the Green Guard system into iterative development cycles. Each sprint contains selected user stories with defined goals and estimated story points to ensure balanced workload distribution and incremental system development.

Sprint Planning Table

Sprint No.	Duration	Sprint Goal	Total Story Points
Sprint 1	2 Weeks	Implement user authentication system US-01, US-02	6
Sprint 2	2 Weeks	Password recovery and profile management US-03, US-04	5
Sprint 3	2 Weeks	Image upload and preprocessing pipeline US-05	5
Sprint 4	2 Weeks	Multimodal dataset synchronization US-06	8
Sprint 5	2 Weeks	CNN-based feature extraction US-07	8
Sprint 6	2 Weeks	ANN-based environmental data processing US-08	5
Sprint 7	2 Weeks	RNN-based temporal analysis US-09	8
Sprint 8	2 Weeks	Multimodal feature fusion US-10	8
Sprint 9	2 Weeks	Ensemble learning module US-11	8
Sprint 10	2 Weeks	Plant disease, nutrient & stress detection US-12, US-13, US-14	15
Sprint 11	2 Weeks	Recommendation engine US-15	5

Sprint 12	2 Weeks	Backend API integration US-16	8
Sprint 13	2 Weeks	Evaluation and confidence scoring US-17, US-18	6
Sprint 14	2 Weeks	Explainable AI visualization US-19	5
Sprint 15	2 Weeks	Dashboard and visual analytics US-20, US-21	8
Sprint 16	2 Weeks	Alert and notification system US-22	5
Sprint 17	2 Weeks	Security and authentication hardening US-20, US-21	10
Sprint 18	2 Weeks	Deployment setup US-22	8
Sprint 19	2 Weeks	Bug fixing and optimization	5
Sprint 20	2 Weeks	Final testing and documentation	3

3. Functional Requirements

3.1 Use Case Diagram

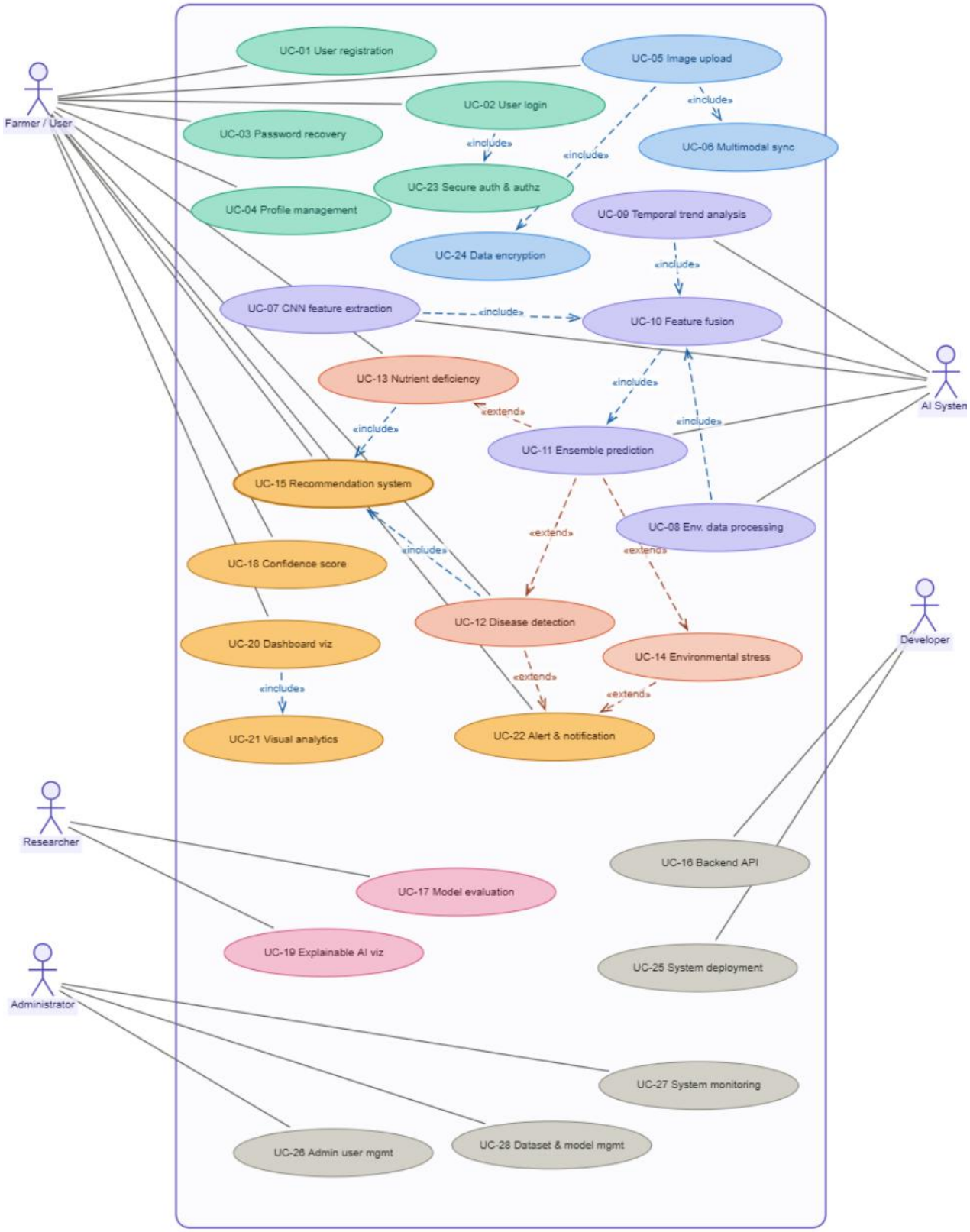


Figure 11: Diagram showing breakdown of Green Guard into major use cases

Use Case Diagram Table

Use Case ID	Use Case Name	Primary Actor	Description	Mapped User Story ID	User Story
UC-01	User Registration	New User	Allows a new user to create an account using email and password	US-01	As a user, I want to register an account so that I can access the system
UC-02	User Login	Registered User	Enables secure login to access dashboard and system features	US-02	As a user, I want to log in so that I can securely access my dashboard
UC-03	Password Recovery	User	Allows users to reset password via email verification	US-03	As a user, I want to reset my password so that I can recover my account
UC-04	Profile Management	Registered User	Allows users to view and update profile information	US-04	As a user, I want to manage my profile so that I can update my personal information
UC-05	Image Upload & Preprocessing	Farmer/User	Enables upload and preprocessing of plant images for analysis	US-05	As a user, I want to upload plant images so that the system can analyze plant health
UC-06	Multimodal Data Synchronization	System	Synchronizes image, environmental, and temporal data	US-06	As a system, I want to synchronize multimodal datasets so that analysis can be performed accurately
UC-07	CNN Feature Extraction	AI System	Extracts visual features from	US-07	As a system, I want to extract visual features so that diseases can

			plant images using CNN		be identified accurately
UC-08	Environmental Data Processing	AI System	Processes environmental data using ANN	US-08	As a system, I want to process environmental data so that crop conditions can be evaluated
UC-09	Temporal Trend Analysis	AI System	Analyzes sequential data using RNN	US-09	As a system, I want to analyze temporal trends so that crop behavior can be understood over time
UC-10	Feature Fusion	AI System	Combines CNN, ANN, and RNN outputs into unified representation	US-10	As a system, I want to combine multiple data sources so that I can make better predictions
UC-11	Ensemble Learning Prediction	AI System	Combines multiple model outputs for improved accuracy	US-11	As a system, I want to combine model outputs so that predictions become more reliable
UC-12	Disease Detection	Farmer/User	Detects plant diseases with confidence score	US-12	As a user, I want to detect plant diseases so that I can take timely action
UC-13	Nutrient Deficiency Detection	Farmer/User	Identifies nutrient deficiencies in crops	US-13	As a user, I want to detect nutrient deficiencies so that I can improve crop health
UC-14	Environmental Stress Detection	Farmer/User	Detects stress conditions in crops	US-14	As a user, I want to detect environmental stress so that I can protect crops from damage

UC-15	Recommendation System	Farmer/User	Generates crop recommendations based on predictions	US-15	As a user, I want recommendations so that I can improve crop yield
UC-16	Backend API Integration	System/Developer	Enables communication between frontend and AI models	US-16	As a system, I want APIs so that frontend and models can communicate
UC-17	Model Evaluation	Researcher	Generates performance metrics (accuracy, precision, recall, F1-score)	US-17	As a researcher, I want evaluation results so that I can measure model performance
UC-18	Confidence Score Display	User	Displays prediction confidence values	US-18	As a user, I want to view prediction confidence so that I can trust results
UC-19	Explainable AI Visualization	Researcher	Provides interpretability using visual explanations	US-19	As a researcher, I want explainable AI so that I can interpret model decisions
UC-20	Dashboard Visualization	User	Displays predictions and analytics in a dashboard	US-20	As a user, I want a dashboard so that I can view all system outputs in one place
UC-21	Visual Analytics	User	Provides graphs and trend visualization	US-21	As a user, I want visual analytics so that I can understand trends easily
UC-22	Alert & Notification System	User	Sends alerts for crop risks	US-22	As a user, I want alerts so that I can be notified of crop risks immediately

UC-23	Secure Authentication & Authorization	User/System	Ensures secure login and access control	US-24	As a user, I want secure authentication so that my data remains protected
UC-24	Data Encryption	System	Encrypts sensitive data for security	US-25	As a system, I want data encryption so that privacy is maintained
UC-25	System Deployment	Developer	Deploys system in production environment	US-23	As a developer, I want deployment setup so that system can run in production
UC-26	Administrator User Management	Administrator	Manages users, roles, and permissions	US-01	As an admin, I want to manage users so that system access is controlled properly
UC-27	System Monitoring & Logs	Administrator	Monitors system performance and logs	US-24	As an admin, I want to monitor system activity so that issues can be detected early
UC-28	Dataset & Model Management	Administrator	Manages datasets and model updates	US-06	As an admin, I want to manage datasets/models so that system stays updated

4. Non-functional Requirements

4.1 Reliability

Green Guard ensures reliable detection and alert generation by maintaining consistent uptime and stable AI model performance. The system is designed to handle continuous user requests without failure, even under heavy usage. Following table shows the reliability breakdown:

Requirement ID	Requirement Description	Related User Stories
NFR-R1	The system should provide stable and consistent prediction results during continuous operation.	US-12, US-13, US-14
NFR-R2	The system should maintain accurate synchronization of image, environmental, and temporal datasets before analysis.	US-06
NFR-R3	The system should recover gracefully from invalid uploads, failed requests, or interrupted operations.	US-05, US-16
NFR-R4	The system should ensure reliable communication between frontend modules and backend AI services.	US-16
NFR-R5	The system should securely maintain active user sessions during system usage.	US-02, US-24

4.2 Performance

The system is optimized for fast response time in processing web inputs and generating disease or stress predictions. Efficient backend processing and lightweight API design ensure minimal delay between user action and results. This supports near real-time monitoring for users.

Requirement ID	Requirement Description	Related User Stories
NFR-P1	The system should process uploaded plant images within acceptable response time limits.	US-05, US-12
NFR-P2	The system should generate prediction results efficiently	US-10, US-11

	after receiving multimodal input data.	
NFR-P3	Backend APIs should return prediction responses with low latency.	US-16
NFR-P4	The dashboard should load analytics, predictions, and graphs without significant delay.	US-20, US-21
NFR-P5	The system should support simultaneous user access without major performance degradation.	US-01, US-02, US-20

4.3 Use-ability

Green Guard provides a simple and intuitive web interface so users can easily upload or input data and view results. Clear dashboards and visual feedback make it accessible even for non-technical users. The design focuses on ease of navigation and quick understanding of plant health status.

Requirement ID	**Requirement Description**	**Related User Stories**
NFR-U1	The system should provide a simple and user friendly interface for all users.	US-01, US-20
NFR-U2	Users should be able to upload plant images and access prediction results easily.	US-05, US-12
NFR-U3	Prediction outputs should be displayed in a clear and understandable format.	US-18, US-20
NFR-U4	The system should provide visual analytics and confidence scores for better understanding.	US-18, US-21
NFR-U5	Error messages and validation alerts shall be understandable and informative.	US-01, US-02, US-05

4.4 Other Non-functional Requirements

4.4.1 Security Requirements

The system ensures secure user authentication, encrypted data storage, and protected API access to safeguard sensitive agricultural and user information.

Requirement ID	**Requirement Description**	**Related User Stories**
NFR-S1	The system should provide secure authentication and authorization mechanisms.	US-02, US-24
NFR-S2	Sensitive user data should be encrypted during storage and transmission.	US-25
NFR-S3	Unauthorized access attempts should be detected and restricted.	US-24
NFR-S4	User sessions should remain protected during active usage.	US-24

4.4.2 Scalability Requirements

Green Guard is designed to handle increasing numbers of users and plant data efficiently by supporting modular cloud based expansion without performance degradation.

Requirement ID	**Requirement Description**	**Related User Stories**
NFR-SC1	The system architecture should support future integration of additional datasets and AI models.	US-06, US-10
NFR-SC2	The system should support increasing numbers of users and agricultural datasets.	US-20
NFR-SC3	The framework should support future integration of additional sensor modalities.	US-08, US-09

4.4.3 Explainability & Validation Requirements

Green Guard provides interpretable AI outputs, allowing users to understand disease detection results and recommendations with clear, human readable insights.

Requirement ID	Requirement Description	Related User Stories
NFR-E1	The system should provide explainable AI visualizations for prediction interpretation.	US-19
NFR-E2	Prediction confidence scores should be displayed with outputs to improve user trust.	US-18
NFR-E3	Biological and agricultural knowledge should be used only for dataset interpretation and output validation purposes.	US-12, US-13, US-14
NFR-E4	All system analysis and prediction operations should remain computational and AI-driven.	US-07, US-08, US-09, US-10

5. Scrum Activities Plan

The Green Guard project adopts the Scrum framework under the Agile methodology to ensure iterative development, continuous feedback, and efficient collaboration among team members. Scrum activities are structured to support the distributed responsibilities of the team and the 2 week sprint cycle defined for the project.

1. Daily Scrum Activities

Daily Scrum activities are conducted to ensure continuous progress tracking and coordination among team members. Due to the academic schedule and distributed working environment, a hybrid communication approach is adopted.

Approach:

- The team maintains daily progress updates through a dedicated communication channel
- Updates are also reflected in the Notion Scrum board for transparency and tracking.
- Each team member provides a structured update including:
    - Tasks completed since the last update.
    - Planned tasks for the current day.
    - Any technical or implementation blockers.

Frequency:

- Daily
- Short synchronization meetings: 2 times per sprint (10–15 minutes) when required

Objective:

- Ensure continuous visibility of project progress
- Identify and resolve blockers early
- Maintain alignment across development tasks

2. Sprint Review Process

Sprint Review meetings are conducted at the every 2 week sprint to evaluate completed work and demonstrate functional deliverables.

Approach:

- Each team member presents completed sprint tasks based on assigned responsibilities:
    - Backend and frontend modules
    - AI/ML model development and integration
    - Data preprocessing and evaluation components
- Functional demonstrations are conducted using GitHub repositories, or Notion documentation updates.

Participants:

- Development Team
- Product Owner

Activities:

- Presentation of completed user stories.
- Verification of acceptance criteria.
- Review of implemented features.
- Collection of feedback from the Product Owner.

Outcome:

- Approval or revision of completed sprint deliverables.
- Updates to product backlog if required.
- Identification of improvements for future sprints.

3. Sprint Retrospective Process

Sprint Retrospective sessions are conducted after each Sprint Review to reflect on the development process and improve future performance.

Approach:

- Internal team meeting conducted
- Findings are documented in the Notion workspace under “Sprint Retrospective Notes”

Activities:

Each retrospective focuses on the following aspects:

- Successful practices during the sprint.
- Technical or coordination challenges faced.
- Areas requiring improvement.

Key Focus Areas:

- Optimization of AI model training and performance.
- Improvement in dataset preprocessing and integration.
- Enhancement of frontend backend communication.
- Better workload distribution and time management.

Outcome:

- Continuous process improvement.
- Increased team efficiency across sprints.

5.1 Kanban Board / Task Tracking

The Green Guard project follows an Agile based task tracking system using a Kanban board implemented in Notion. This ensures transparency, continuous progress monitoring, and effective collaboration among team members.

Task Tracking Methodology:

All project tasks are managed using a Notion Kanban board, which provides a visual workflow for tracking development progress across all modules of the system.

The Kanban system is divided into the following task states:

- **To Do:** Tasks that are planned and not yet started.
- **In Progress:** Tasks currently being actively developed.
- **Testing:** Tasks under validation, evaluation, or debugging.
- **Done:** Fully completed and verified tasks.

Workflow Structure:

The project follows a hybrid task assignment model, where:

- Tasks are assigned based on module ownership (Frontend, Backend, AI/ML)
- Team members can assist each other across modules when required.
- This ensures flexibility while maintaining accountability.

Tool Used:

The following tools are used for task tracking and collaboration:

- Notion: Primary tool for task management and progress tracking.
- GitHub: Used for code version control and integration tracking.

The Notion board is accessible to all team members and is also shared with the supervisor for real time project monitoring and evaluation.

Kanban Board:

Sprint wise Kanban Board:

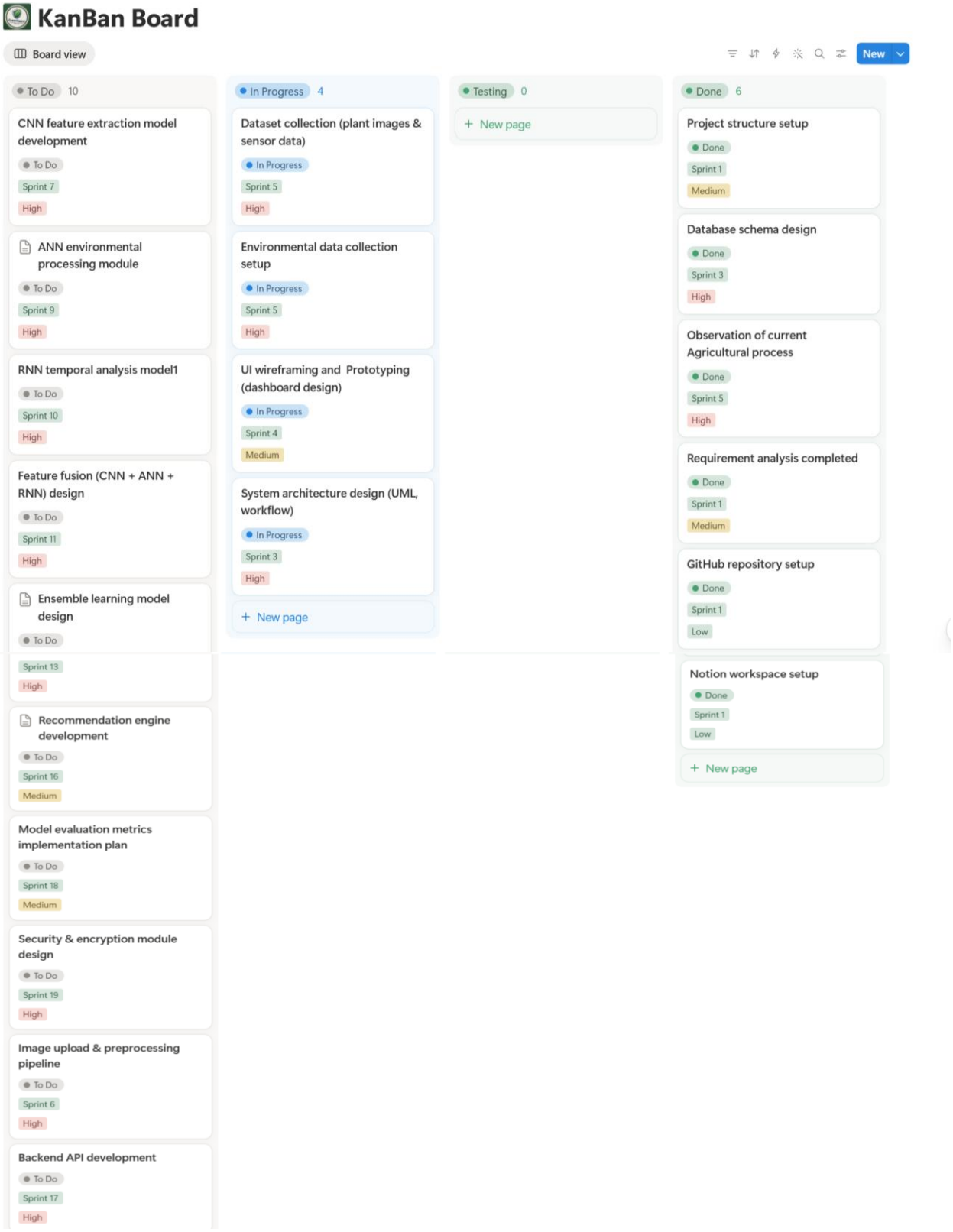


Figure 12: Sprint Wise Notion Kanban Board

Task wise Kanban Board:

To Do	In Progress	Testing	Done
CNN feature extraction model development	Dataset collection (plant images & sensor data)	Basic dataset sanity checks (missing values, duplicates removal)	Project structure setup
ANN environmental processing module	Environmental data collection setup	Initial image quality validation (blur/invalid images filtering)	Database schema design
RNN temporal analysis model	System architecture design (UML, workflow)	API response format validation (basic endpoints)	User authentication system
Feature fusion (CNN + ANN + RNN) design	UI wireframing and Prototyping (dashboard design)	Model input shape validation (CNN/ANN/RNN compatibility checks)	Requirement analysis completed
Ensemble learning model design			GitHub repository setup
Recommendation engine development			Notion workspace setup
Model evaluation metrics implementation plan			Initial dataset cleaning & labeling
Security & encryption module design			
Image upload & preprocessing pipeline			
Backend API development			

5.2 Acceptance Criteria

Feature	**Acceptance Conditions**	**Test Scenarios**	**Expected Results**
User Registration & Login	Valid users can register/login; invalid inputs rejected	Register new user, login with wrong credentials	Account created successfully; error shown for invalid login
Password Recovery	Email based reset link works securely	Request password reset	Reset link sent; password updated successfully
Image Upload & Preprocessing	Supports valid image formats and preprocessing pipeline	Upload JPG/PNG, upload invalid file	Valid images processed; invalid files rejected
Multimodal Data Sync	Image, environmental, and temporal data aligned correctly	Upload full/partial multimodal data	Unified dataset created; error if missing data
CNN Disease Detection	CNN correctly classifies plant diseases	Upload diseased/healthy leaf image	Correct label with confidence score
ANN Environmental Analysis	Processes sensor data accurately	Normal vs extreme environmental input	Correct environmental feature output
RNN Temporal Analysis	Captures time series trends correctly	Stable vs fluctuating historical data	Accurate trend prediction generated
Feature Fusion (CNN+ANN+RNN)	Combined features form unified representation	Full multimodal input test	Single optimized feature vector produced
Ensemble Prediction	Final model improves accuracy via ensemble	Compare single vs ensemble output	More stable and accurate prediction

Recommendation System	Generates valid agricultural advice	Disease/nutrient deficiency cases	Relevant recommendations shown
Dashboard & Visualization	Displays analytics without UI errors	Load dashboard with data	Charts, graphs, predictions displayed correctly
Alert System	Generates real time alerts for risks	Disease/stress detection event	Instant alert notification triggered
Security & Authentication	Prevent unauthorized access	Invalid login / API access attempt	Access denied; session secured
Performance & Reliability	System responds within acceptable time	Multiple users, large input data	Stable response, no system crash

5.3 Release Criteria

Area	**Release Condition**	**Validation Method**
Functional Completion	All core modules (authentication, AI prediction, dashboard, alerts) are implemented	User story and functional testing
AI Model Integration	CNN, ANN, RNN, and fusion pipeline work end to end	Integration testing
Prediction Performance	Model meets required accuracy and produces stable outputs	Evaluation metrics (accuracy, precision, recall, F1-score)
Recommendation System	Generates correct and relevant farming advice	Scenario based testing
UI Functionality	Dashboard displays predictions and analytics correctly	UI testing
API Functionality	Backend APIs respond correctly without failure	API testing

Security	Authentication and access control work correctly	Security testing
Performance	System responds within acceptable time limits	Load/performance testing
Deployment	System is successfully deployed on target environment	Deployment verification

5.4 Burndown Chart & Progress Tracking

The Green Guard project uses a sprint based burndown chart to monitor overall project progress. Progress is measured using effort points assigned to each sprint activity including research, design, dataset preparation, development, testing, and deployment.

Work Tracking Method:

- Sprint wise task tracking using Agile Scrum.
- Effort points assigned based on workload and complexity.
- Burndown chart updated after every sprint.
- Tracks completed vs remaining project effort.

Expected Burn Rate:

- Early sprints have moderate burn rate due to research and planning activities.
- Development and integration sprints consume higher effort points.
- A gradual downward trend indicates healthy project progress.

Process Tracking:

Sprint	Planned Points	Points Completed	Remaining Points	Velocity	Status
Sprint 1	8	8	140	8	Completed
Sprint 2	8	8	132	8	Completed
Sprint 3	10	10	122	10	Completed
Sprint 4	6	6	116	6	Completed
Sprint 5	8	8	108	8	Completed
Sprint 6	8	0	108		Active
Sprint 7	10	0	100		Upcoming
Sprint 8	8	0	92		Upcoming

Sprint 9	8	0	84		Upcoming
Sprint 10	10	0	74		Upcoming
Sprint 11	8	0	66		Upcoming
Sprint 12	8	0	58		Upcoming
Sprint 13	6	0	52		Upcoming
Sprint 14	6	0	46		Upcoming
Sprint 15	5	0	41		Upcoming
Sprint 16	8	0	33		Upcoming
Sprint 17	8	0	25		Upcoming
Sprint 18	6	0	19		Upcoming
Sprint 19	5	0	14		Upcoming
Sprint 20	4	0	10		Upcoming

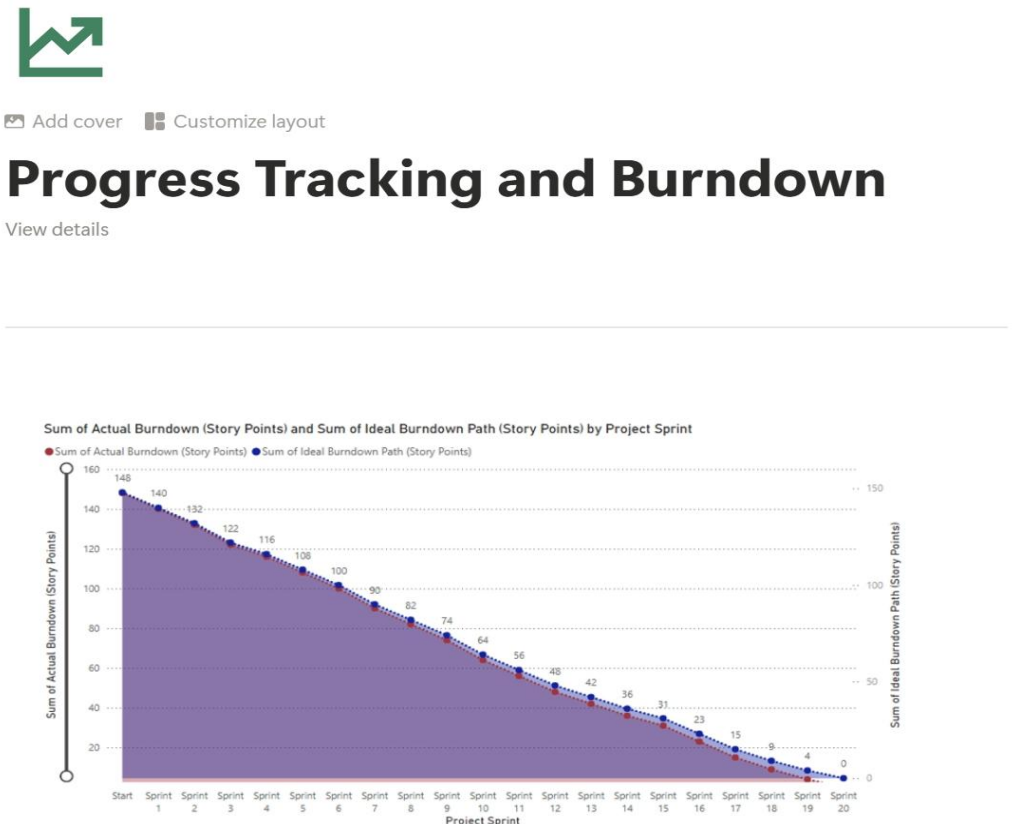


Figure 13: Notion Burndown Chart of Green Guard

5.5 Backlog Refinement Plan

Backlog refinement in Green Guard is a sprint based activity used to keep the product backlog updated, prioritized, and aligned with ongoing development of the AI based plant monitoring system.

Frequency:

- Conducted once per sprint (every 2 weeks).
- Updated after each sprint review based on feedback.
- Minor updates can be made anytime in Notion.

Process:

- **Reprioritize tasks** based on sprint progress and AI pipeline needs.
- **Refine user stories** by adding missing details and acceptance criteria.
- **Adjust story points** based on actual development effort.
- **Split large tasks** into smaller sprint ready tasks.
- **Update backlog** using sprint results, model performance, and supervisor feedback.
- **Maintain tracking** using Notion and GitHub commits.

6. Risk Analysis

Risk analysis helps identify potential challenges that may affect the successful development and performance of the Green Guard system. The following technical, operational, and project related risks were identified along with their possible impacts and mitigation strategies.

Risk ID	Risk Description	Impact	Mitigation Strategy
R-01	Limited availability of fully synchronized multimodal agricultural datasets	May affect training consistency and prediction quality	Use multiple publicly available datasets and apply standardized preprocessing techniques
R-02	Poor quality or noisy plant images	May reduce disease detection accuracy	Apply image preprocessing and data augmentation techniques
R-03	High computational requirements during model training	May increase training time and resource usage	Use lightweight pretrained deep learning models and cloud-based GPU resources
R-04	Difficulty in integrating multiple AI models (CNN, ANN, RNN)	May delay system integration and testing	Develop and test each module independently before final integration
R-05	Overfitting due to limited dataset diversity	May reduce real-world prediction performance	Use regularization, dropout, and validation techniques
R-06	Backend or API communication failures	May interrupt prediction and dashboard services	Implement proper API testing and error handling mechanisms
R-07	Security and privacy risks related to user data	May affect system trust and data protection	Use secure authentication and encryption techniques
R-08	Time management and workload challenges during development	May affect timely completion of project tasks	Divide responsibilities among members of

			the team s and follow a planned development schedule

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Appendix A: Glossary

Term / Acronym	*Abbreviation*	*Definition*
A		
Artificial Intelligence	*AI*	*Computer systems performing tasks requiring human-like intelligence.*
Artificial Neural Network	*ANN*	*Computational model inspired by biological neural networks.*
Authentication		*Verifying a user's identity before system access.*
Authorization		*Granting system access based on user roles and privileges.*
C		
Cloud Deployment		*Hosting system components on cloud infrastructure for scalability.*
Convolutional Neural Network	*CNN*	*Deep learning model for image analysis, e.g., plant disease detection.*
D		
Dashboard		*Graphical interface showing plant health, sensor readings, and alerts.*
Dataset		*Structured collection of images and sensor data for model training.*
Deep Learning	*DL*	*Multi-layer neural networks for modelling complex patterns in data.*
Decision-Support System		*System providing insights to assist user decision making.*
E		
Ensemble Learning		*Combining multiple models to improve prediction accuracy.*
G		
Green Guard		*Web based intelligent system for plant health monitoring and alerts.*
I		
Inference		*Generating predictions from a trained machine learning model.*
Internet of Things	*IOT*	*Network of sensors and devices transmitting real-time environmental data.*
M		
Machine Learning	*ML*	*Systems learning patterns from data without explicit programming.*
Multi-Stress Analysis		*Detecting diseases, deficiencies, and stress simultaneously in plants.*
R		

Recurrent Neural Network	*RNN*	*Neural network for sequential or time-series data processing.*
S		
Sensor Data		*Environmental measurements collected from IoT sensors.*
SoftwareRequirements Specification	*SRS*	*Document defining functional and non-functional requirements.*
U		
User Role		*Defined access level assigned to a system user.*
User Interface/User Experience	*UI/UX*	*Design principles for user interface and user experience.*

Appendix B: IV & V Report

(Independent verification & validation)
IV & V Resource

Name Signature

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
…					

Table 1: List of non-trivial defects